

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

Slide Set 10: Validation



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
 - ▶ For use only by enrolled students in the course

So, how to pick λ , the regularization strength?

Hyper-Parameters

So, how to pick λ , the regularization strength?

λ is an example of a **hyper-parameter**:

- Hyper-parameters: parameters of the learning algorithm (that picks the parameters, $\vec{w}$).
- Cannot pick hyper-parameters based on the data used to pick the parameters

Hyper-Parameters

So, how to pick λ , the regularization strength?

λ is an example of a **hyper-parameter**:

- Hyper-parameters: parameters of the learning algorithm (that picks the parameters, $\vec{w}$).
- Cannot pick hyper-parameters based on the data used to pick the parameters

We will use a validation set: data separate from the data used to pick the parameters
But, **not** the testing data!

Hold-out Validation

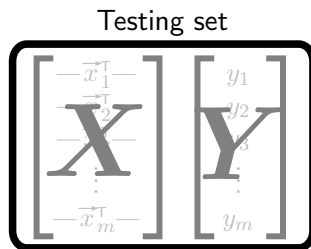
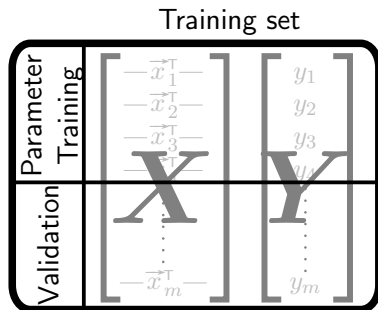
Training set

Parameter Training	$\begin{bmatrix} -\vec{x}_1^\top - \\ -\vec{x}_2^\top - \\ -\vec{x}_3^\top - \\ \vdots \\ -\vec{x}_m^\top - \end{bmatrix}$	$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}$
Validation	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}$

Testing set

$\begin{bmatrix} -\vec{x}_1^\top - \\ -\vec{x}_2^\top - \\ -\vec{x}_3^\top - \\ \vdots \\ -\vec{x}_m^\top - \end{bmatrix}$	$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}$
--	--

Hold-out Validation



$(\text{ptrain}, \text{valid}) \leftarrow \text{Split}(\text{training})$

$\text{besterr} \leftarrow \infty$

for all $\gamma \in \Gamma$ **do**

$\vec{w} \leftarrow \text{Learn}(\text{ptrain}, \gamma)$

$\text{verr} \leftarrow \text{Error}(\text{valid}, \vec{w})$

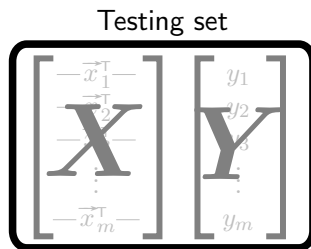
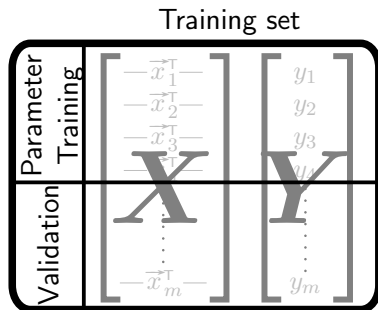
if $\text{verr} < \text{besterr}$ **then**

$\text{besterr} \leftarrow \text{verr}$

return $\gamma^* \leftarrow \gamma$

Γ is the set of hyper-parameters to try

Hold-out Validation



$(\text{ptrain}, \text{valid}) \leftarrow \text{Split}(\text{training})$

$\text{besterr} \leftarrow \infty$

for all $\gamma \in \Gamma$ **do**

$\vec{w} \leftarrow \text{Learn}(\text{ptrain}, \gamma)$

$\text{verr} \leftarrow \text{Error}(\text{valid}, \vec{w})$

if $\text{verr} < \text{besterr}$ **then**

$\text{besterr} \leftarrow \text{verr}$

return $\gamma^* \leftarrow \gamma$

Γ is the set of hyper-parameters to try

Testing set never used!

- Can be used to pick any hyper-parameter, or hyper-parameters
 - ▶ examples: degree of polynomial, regularization strength, type of regularization
 - ▶ If picking multiple, combinational explosion can cause problems

- Can be used to pick any hyper-parameter, or hyper-parameters
 - ▶ examples: degree of polynomial, regularization strength, type of regularization
 - ▶ If picking multiple, combinational explosion can cause problems
- In classification:
 - ▶ Error used for validation is number incorrect (not per-item error used in learning)
 - ▶ If class imbalance, use stratified partitioning

- Can be used to pick any hyper-parameter, or hyper-parameters
 - ▶ examples: degree of polynomial, regularization strength, type of regularization
 - ▶ If picking multiple, combinational explosion can cause problems
- In classification:
 - ▶ Error used for validation is number incorrect (not per-item error used in learning)
 - ▶ If class imbalance, use stratified partitioning
- How to pick $\vec{w}$
 - ▶ After hyper-parameter has been picked,
 - ▶ ... Re-run learning using this hyper-parameter,
 - ▶ ... On all the training data

Cross Validation

Training set

fold 1	$\begin{bmatrix} -\vec{x}_1^T- \\ -\vec{x}_2^T- \\ -\vec{x}_3^T- \\ \vdots \\ -\vec{x}_m^T- \end{bmatrix}$	$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}$
fold 2		
fold 3		
$\vdots$		
fold k		

Testing set

$\begin{bmatrix} -\vec{x}_1^T- \\ -\vec{x}_2^T- \\ -\vec{x}_3^T- \\ \vdots \\ -\vec{x}_m^T- \end{bmatrix}$	$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}$
--	--

Cross Validation

Training set

fold 1	$\begin{bmatrix} -\vec{x}_1^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_1 \\ \vdots \\ \end{bmatrix}$
fold 2	$\begin{bmatrix} -\vec{x}_2^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_2 \\ \vdots \\ \end{bmatrix}$
fold 3	$\begin{bmatrix} -\vec{x}_3^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_3 \\ \vdots \\ \end{bmatrix}$
$\vdots$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \end{bmatrix}$
fold k	$\begin{bmatrix} -\vec{x}_m^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_m \\ \vdots \\ \end{bmatrix}$

Testing set

$\begin{bmatrix} -\vec{x}_1^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_1 \\ \vdots \\ \end{bmatrix}$
$\begin{bmatrix} -\vec{x}_2^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_2 \\ \vdots \\ \end{bmatrix}$
$\begin{bmatrix} -\vec{x}_3^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_3 \\ \vdots \\ \end{bmatrix}$
$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \end{bmatrix}$
$\begin{bmatrix} -\vec{x}_m^\top - \\ \vdots \\ \end{bmatrix}$	$\begin{bmatrix} y_m \\ \vdots \\ \end{bmatrix}$

$(f_1, \dots, f_k) \leftarrow \text{Split-k}(\text{training})$

$\text{besterr} \leftarrow \infty$

for all $\gamma \in \Gamma$ **do**

$\text{verr} \leftarrow 0$

for all $j \in \{1, 2, \dots, k\}$ **do**

$\vec{w} \leftarrow \text{Learn}(\text{training} - f_j, \gamma)$

$\text{verr} \leftarrow \text{verr} + \text{Error}(f_j, \vec{w})$

if $\text{verr} < \text{besterr}$ **then**

$\text{besterr} \leftarrow \text{verr}$

$\gamma^* \leftarrow \gamma$

return γ^*

Γ is the set of hyper-parameters to try

Cross Validation

Training set

fold 1	$\begin{bmatrix} -\vec{x}_1^T - \end{bmatrix}$	$\begin{bmatrix} y_1 \end{bmatrix}$
fold 2	$\begin{bmatrix} -\vec{x}_2^T - \end{bmatrix}$	$\begin{bmatrix} y_2 \end{bmatrix}$
fold 3	$\begin{bmatrix} -\vec{x}_3^T - \end{bmatrix}$	$\begin{bmatrix} y_3 \end{bmatrix}$
$\vdots$	$\begin{bmatrix} \mathbf{X} \end{bmatrix}$	$\begin{bmatrix} \mathbf{Y} \end{bmatrix}$
fold k	$\begin{bmatrix} -\vec{x}_m^T - \end{bmatrix}$	$\begin{bmatrix} y_m \end{bmatrix}$

Testing set

$\begin{bmatrix} -\vec{x}_1^T - \end{bmatrix}$	$\begin{bmatrix} y_1 \end{bmatrix}$
$\begin{bmatrix} -\vec{x}_2^T - \end{bmatrix}$	$\begin{bmatrix} y_2 \end{bmatrix}$
$\begin{bmatrix} -\vec{x}_3^T - \end{bmatrix}$	$\begin{bmatrix} y_3 \end{bmatrix}$
$\begin{bmatrix} \mathbf{X} \end{bmatrix}$	$\begin{bmatrix} \mathbf{Y} \end{bmatrix}$
$\begin{bmatrix} -\vec{x}_m^T - \end{bmatrix}$	$\begin{bmatrix} y_m \end{bmatrix}$

$(f_1, \dots, f_k) \leftarrow \text{Split-k}(\text{training})$

$\text{besterr} \leftarrow \infty$

for all $\gamma \in \Gamma$ **do**

$\text{verr} \leftarrow 0$

for all $j \in \{1, 2, \dots, k\}$ **do**

$\vec{w} \leftarrow \text{Learn}(\text{training} - f_j, \gamma)$

$\text{verr} \leftarrow \text{verr} + \text{Error}(f_j, \vec{w})$

if $\text{verr} < \text{besterr}$ **then**

$\text{besterr} \leftarrow \text{verr}$

$\gamma^* \leftarrow \gamma$

return γ^*

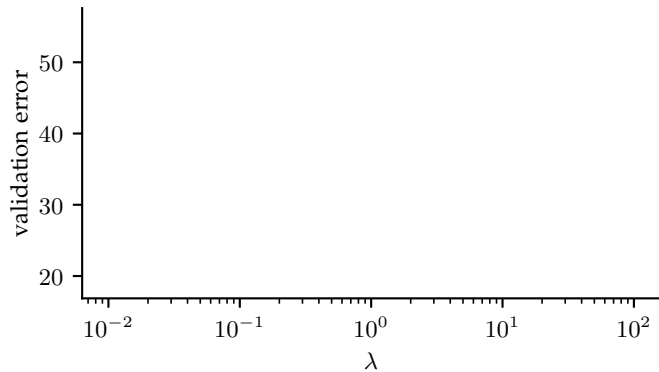
Γ is the set of hyper-parameters to try

Testing set never used!

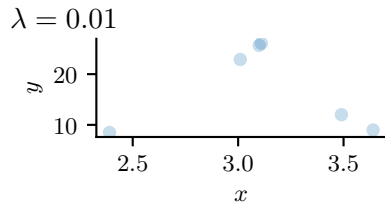
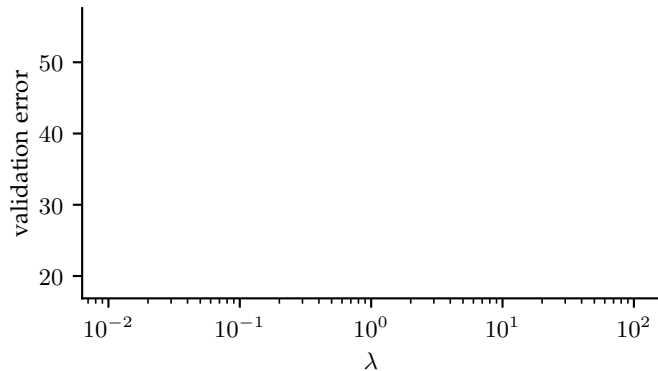
Notes on Cross-Validation

- Called **cross-validation**, **k-fold cross-validation**, or **CV**
- When $k = m$, called **leave-one-out cross-validation** (LOOCV)
- Picking k :
 - ▶ Too small and parameter learning set much smaller than full training set to be used
 - ▶ Too large and overly confident estimates of error
 - ▶ Generally pick $k = 5$ or $k = 10$

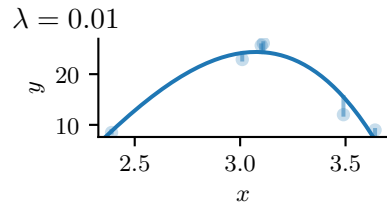
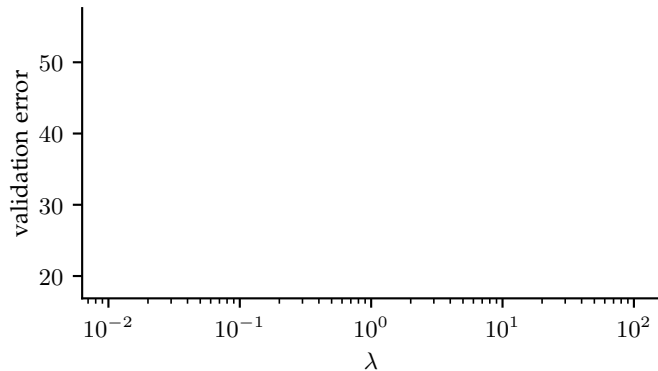
Hold-out Validation Example



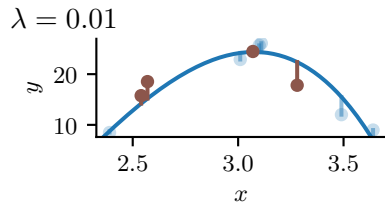
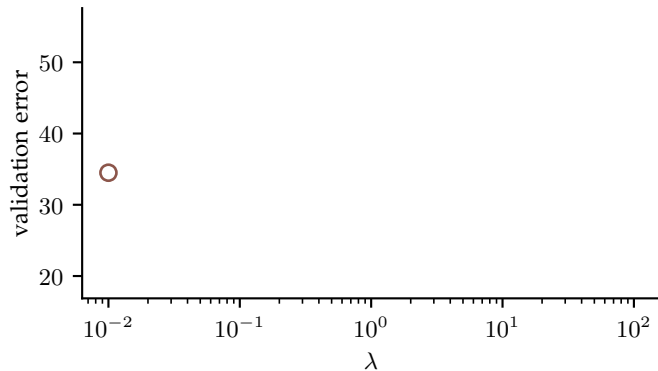
Hold-out Validation Example



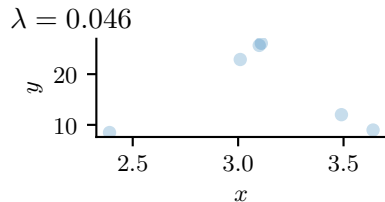
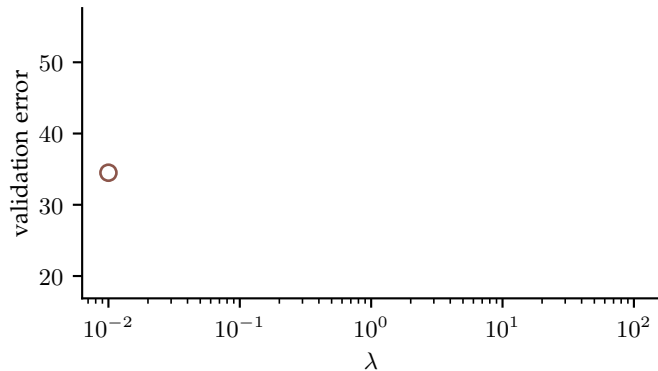
Hold-out Validation Example



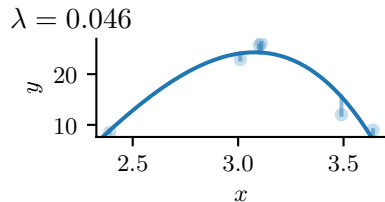
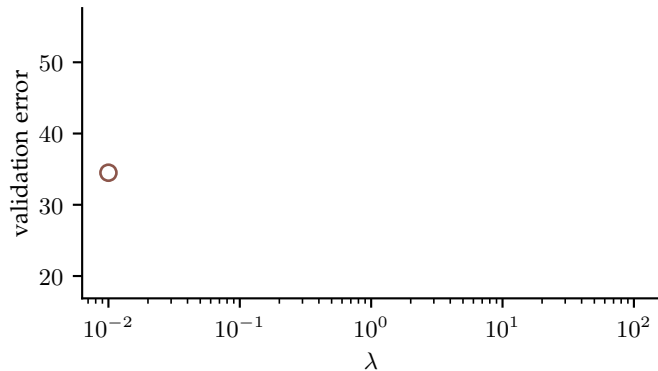
Hold-out Validation Example



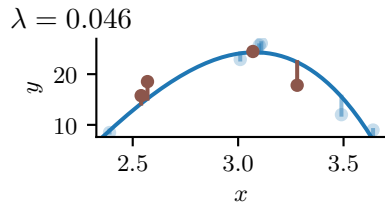
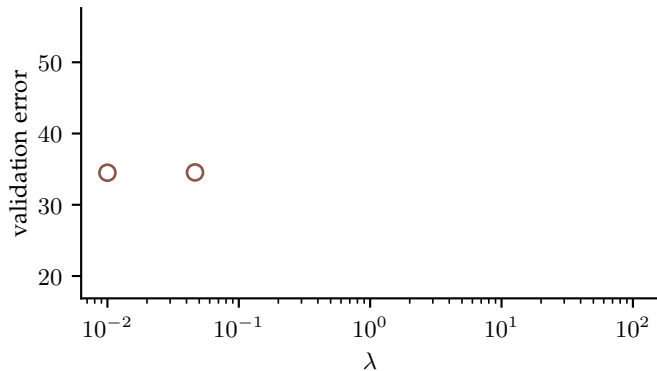
Hold-out Validation Example



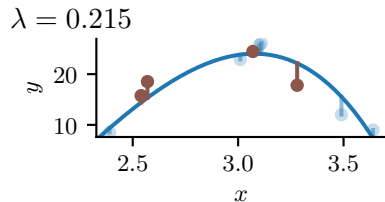
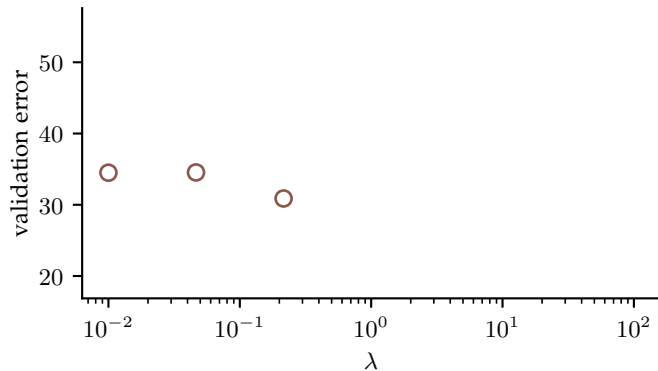
Hold-out Validation Example



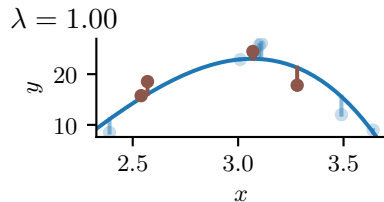
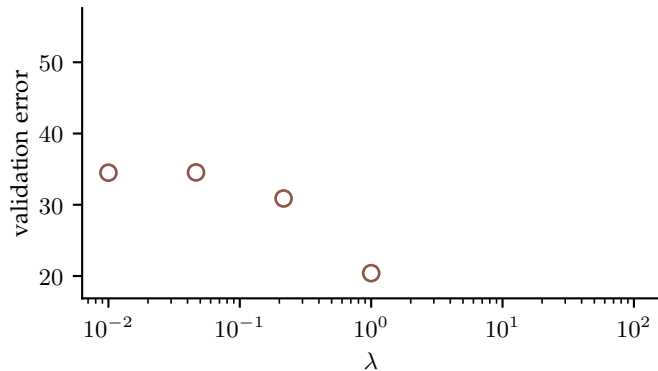
Hold-out Validation Example



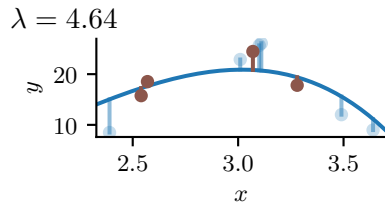
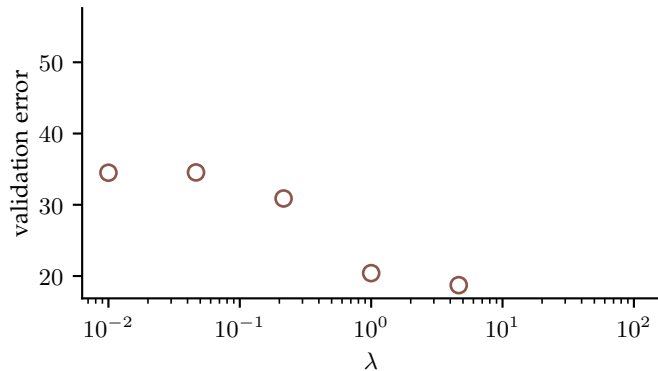
Hold-out Validation Example



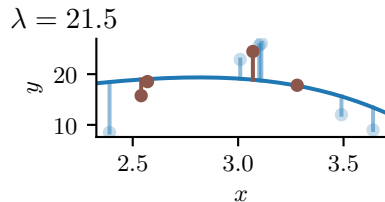
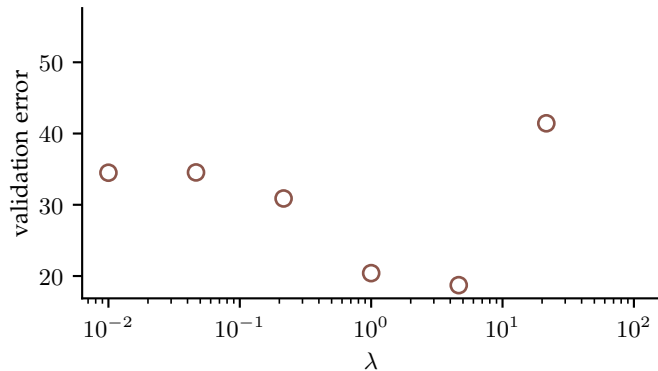
Hold-out Validation Example



Hold-out Validation Example



Hold-out Validation Example



Hold-out Validation Example

